

MySQL Assignment 1 - DDL Commands, Constraints

Problem Statement:

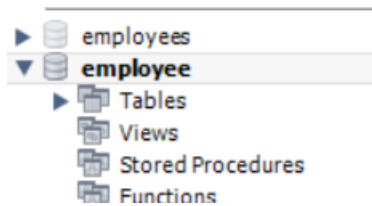
As a Data Analyst, the company requires an Employee Database to store and manage information

related to employees, departments, and locations within the organization.

You are required to design and manage this employee database using MySQL DDL commands, focusing on creating, modifying, and managing database objects while enforcing appropriate

DDL Commands

1.Database and Table Creation (CREATE):



2. Table Alteration (ALTER)

```

29
30 • ALTER TABLE Employees
31   ADD email VARCHAR(100);
32 • DESC Employees;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Field	Type	Null	Key	Default	Extra
▶	employee_id	int	NO	PRI	NULL	
	employee_name	varchar(100)	YES		NULL	
	gender	char(1)	YES		NULL	
	date_of_joining	date	YES		NULL	
	designation	varchar(50)	YES		NULL	
	salary	decimal(10,2)	YES		NULL	
	department_id	int	YES	MUL	NULL	
	location_id	int	YES	MUL	NULL	
	email	varchar(100)	YES		NULL	

Result 1 x Read Only

3. Table Renaming (RENAME):

```

47 RENAME COLUMN hire_date TO date_of_joining;
48 • DESC Employees;
49 • RENAME TABLE Departments TO Departments_Info;
50 • Rename TABLE Location to Locations;
51 • SHOW TABLES;
52
53 • TRUNCATE TABLE Employees;
54

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Tables_in_employee
departments
employees
location

Result Grid | Form Editor | Field

4. Table Truncation (TRUNCATE):

✓	10:01:19	TRUNCATE TABLE Employees	0 row(s) affected	0.109 sec
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5. Database & Table Dropping (DROP):

✓	5	10:01:19	TRUNCATE TABLE Employees	0 row(s) affected	0.109 sec
✓	6	10:01:33	DROP DATABASE employee	3 row(s) affected	0.078 sec

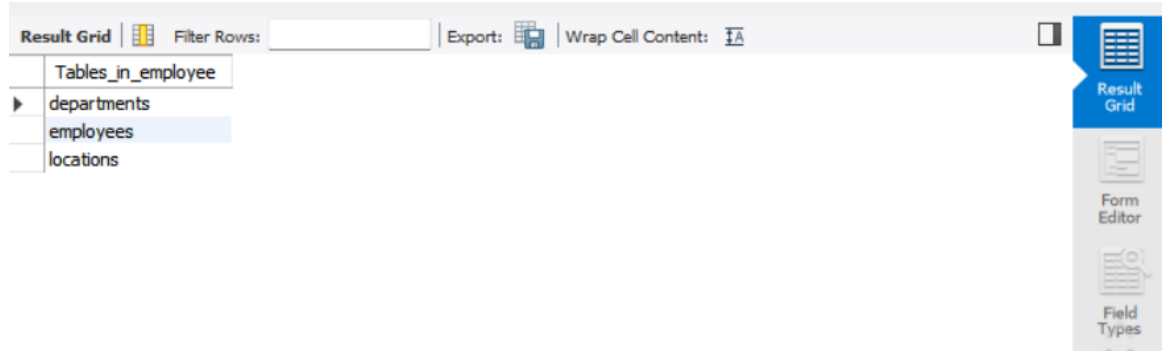
Tasks: Constraints –

1. Database Recreation Ensure that the "department_id" uniquely identifies each department. Set up constraints on the "department name" to avoid duplicate and null entries.
2. Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially. Implement constraints to prevent the insertion of null and duplicate locations.

- `CREATE DATABASE employee;`
- `USE employee;`
- `CREATE TABLE Departments (
 department_id INT PRIMARY KEY,
 department_name VARCHAR(100) NOT NULL UNIQUE
);`
- `CREATE TABLE Locations (
 location_id INT AUTO_INCREMENT PRIMARY KEY,
 location_name VARCHAR(100) NOT NULL UNIQUE
);`

3. Employees Table:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(100)	YES		NULL	
gender	char(1)	YES		NULL	
date_of_joining	date	YES		NULL	
designation	varchar(50)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	



Conclusion:

Successfully created and managed the database by performing DDL operations such as CREATE, ALTER, RENAME, TRUNCATE, and DROP. Tables were created with appropriate constraints, relationships, and foreign keys, demonstrating a clear understanding of database design and management using MySQL.